

Important NP-Complete Problems

1 3-SAT Problem

Definition: 3-SAT

The **3-SAT problem** asks whether a Boolean formula in conjunctive normal form can be made true, where each clause has exactly three literals.

A 3-CNF formula has the form:

$$(C_1) \wedge (C_2) \wedge \cdots \wedge (C_m)$$

Each clause contains exactly three literals.

Example

$$(x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee x_4)$$

Question:

Is there a truth assignment that makes the formula true?

Checking a Certificate

A certificate is a truth assignment such as:

$$x_1 = T, \quad x_2 = T, \quad x_3 = F, \quad x_4 = T$$

We can check every clause in polynomial time. Therefore, 3-SAT is in **NP**.

2 Vertex Cover

Definition: Vertex Cover

Given an undirected graph $G = (V, E)$, a **vertex cover** is a subset $C \subseteq V$ such that every edge in E has at least one endpoint in C .

Decision Version

Given a graph G and an integer k , determine whether G has a vertex cover of size at most k .

$$|C| \leq k$$

Example

Let:

$$V = \{1, 2, 3, 4\}, \quad E = \{(1, 2), (2, 3), (3, 4)\}$$

A vertex cover is:

$$C = \{2, 3\}$$

Because every edge touches vertex 2 or vertex 3.

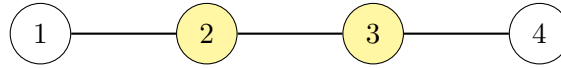


Figure: Vertices 2 and 3 cover all edges.

3 Independent Set

Definition: Independent Set

Given an undirected graph $G = (V, E)$, an **independent set** is a subset $I \subseteq V$ such that no two vertices in I are adjacent.

Decision Version

Given a graph G and an integer k , determine whether G has an independent set of size at least k .

$$|I| \geq k$$

Example

For:

$$V = \{1, 2, 3, 4\}, \quad E = \{(1, 2), (2, 3), (3, 4)\}$$

A valid independent set is:

$$I = \{1, 3\}$$

There is no edge between 1 and 3.

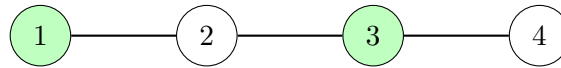


Figure: Vertices 1 and 3 form an independent set.

4 Clique

Definition: Clique

Given an undirected graph $G = (V, E)$, a **clique** is a subset $K \subseteq V$ such that every pair of distinct vertices in K is connected by an edge.

Decision Version

Given a graph G and an integer k , determine whether G contains a clique of size at least k .

$$|K| \geq k$$

Example

Let:

$$V = \{1, 2, 3, 4\}$$

$$E = \{(1, 2), (1, 3), (2, 3), (3, 4)\}$$

Then:

$$K = \{1, 2, 3\}$$

is a clique because every pair among 1, 2, and 3 is connected.

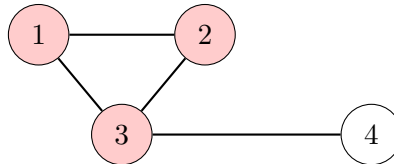


Figure: $\{1, 2, 3\}$ is a clique.

5 Set Cover

Definition: Set Cover

Given a universe:

$$U = \{u_1, u_2, \dots, u_n\}$$

and a collection of subsets:

$$\mathcal{S} = \{S_1, S_2, \dots, S_m\}$$

a **set cover** is a subcollection $\mathcal{C} \subseteq \mathcal{S}$ such that every element of U belongs to at least one set in $\mathcal{C}$.

Decision Version

Given U , $\mathcal{S}$, and integer k , determine whether there exists a set cover of size at most k .

$$|\mathcal{C}| \leq k$$

Example

Let:

$$U = \{1, 2, 3, 4, 5\}$$

$$S_1 = \{1, 2, 3\}, \quad S_2 = \{2, 4\}, \quad S_3 = \{3, 4, 5\}$$

Then:

$$\{S_1, S_3\}$$

is a set cover because:

$$S_1 \cup S_3 = \{1, 2, 3, 4, 5\} = U$$